

1. Introduction

- **Project Overview:** A machine learning-powered web app that predicts credit card approval decisions based on applicant details (age, income, credit score, employment years, etc.).
- **Purpose:** To automate and streamline the approval process, reduce manual errors, and provide instant decisions.

2. Ideation Phase

- **Problem Statement:** Manual credit card approval processes are slow, error-prone, and inconsistent.
- **Empathy Map Canvas:** Applicants want quick, fair, and transparent decisions. Banks want efficiency and reduced risk.
- **Brainstorming:** Explored solutions like rule-based systems, but chose ML for accuracy and scalability.

3. Requirement Analysis

- **Customer Journey Map:** Applicant → Submits details → System validates → Model predicts → Decision displayed.
- **Solution Requirement:** Input validation, secure API, fast response time, clear UI.
- **Data Flow Diagram:** Applicant data → Validation → ML Model → Prediction → UI Output.
- **Technology Stack:** Python (Flask), Bolt Database, ML model (scikit-learn), HTML/CSS/JS frontend, GitHub for version control.

4. Project Design

- **Problem Solution Fit:** ML model provides faster, more accurate approvals.
- **Proposed Solution:** Web app with form inputs, backend ML model, API integration.
- **Solution Architecture:** Frontend (form + UI) → Backend (Flask API) → ML Model → Database → Output.

5. Project Planning & Scheduling

- **Project Planning:** Agile sprints — Sprint 1 (data prep), Sprint 2 (model training), Sprint 3 (API + UI), Sprint 4 (testing & deployment).

6. Functional & Performance Testing

- **Performance Testing:** Response time < 3 seconds, API stable under load, file uploads handled smoothly.
- **Functional Testing:** Input validation, prediction accuracy, API connectivity all passed.

7. Results

- **Output Screenshots:**
 - Applicant form with fields (Name, Age, Income, Credit Score).
 - Prediction result screen showing approval/rejection with probability.

- API status check screen.

8. Advantages & Disadvantages

- **Advantages:** Fast decisions, reduced manual work, scalable, user-friendly.
- **Disadvantages:** Dependent on data quality, potential bias in training data.

9. Conclusion

- The system successfully automates credit card approval predictions with high accuracy and efficiency.

10. Future Scope

- Add explainable AI (show why a decision was made).
- Expand to loan approvals.
- Integrate with mobile apps.

11. Appendix

- **Source-Code:**<https://github.com/rithika1056-collab/AI-ML-PROJECT-CREDIT-CARD-APPROVAL-PREDICTION.git>
- **Dataset Link:** <https://www.kaggle.com/datasets/samueltcortinhas/credit-card-approval-clean-data>